The Pull Bench: worksheet

Name _____ ID _____ 27 points. Use your readings from the bench. Write numbers as fractions. Show the rings, the core and the subtraction.

1. (3 points) Lithium, sodium and potassium are all in group 1. From the bench, fill in a table with a row for each: atom, rings, and the pull as a fraction.

Scale
1 point per atom with the rings and the pull both right

2. (4 points) Look down the column in question 1. Claim: what happens to the pull on the outer electron as you go down a group? Evidence: use two numbers from your table. Reasoning: say what changed about the atom (rings, core, distance).

Scale
Claim states smaller going down: 1
Evidence cites two readings from the table: 1
Reasoning names the extra ring or the greater distance: 1
Reasoning notes that Z_eff did not change: 1

3. (4 points) Before you measured magnesium you picked sodium or magnesium. Write your prediction, the two readings, and whether the data matched. Give one reason magnesium reads the way it does.

Scale
Prediction written before the reading (a wrong prediction earns this point): 1
Both readings correct: 1
Says whether the data matched: 1
Reason uses Z_eff or the extra proton: 1

4. (6 points) After the huddle, work these by hand. For aluminum (Z = 13) and for chlorine (Z = 17): list the rings, count the core electrons, find Z_eff, and find the pull as a fraction.

Scale
Per atom: rings and core right: 1; Z_eff right: 1; pull right: 1 (3 points for each atom)

5. (3 points) The mystery atom on your screen showed rings and a pull. Write its rings, its pull, the atom you named, and how you found it.

Scale
Rings and pull copied right: 1
Atom right: 1
Method adds the rings to find the protons: 1

6. (4 points) Sodium has 11 protons, but its outer electron is pulled as if the nucleus were +1. In two or three sentences, explain how.

Scale
Names the core electrons (inside the outer ring): 1
Says core electrons cancel or shield part of the pull: 1
Gets 11 − 10 = 1: 1
Uses the words Z_eff or effective charge correctly: 1

7. (3 points) Which holds its outer electron harder, sodium or chlorine? Use Z_eff and the ring number for both.

Scale
Chlorine: 1
Z_eff for both (7 and 1): 1
Notes the same outer ring, so the difference is Z_eff: 1